

Core - V: Data Structures

Course Code	Course Name	Category	Hours / Week	Credits
25BCA33C	Data Structures	Core-V	6	4

Course Objectives

The course intends to cover:

- Various data structures algorithms.
- Data representation techniques such as Stack, Queue, List, Trees, Graphs.
- Hash Tables, Hashing Functions
- Sorting and searching methods.

Course Learning Outcomes

On the successful completion of the course, students will be able to

CLO	Content	Knowledge Level
CLO1	Remember the basics of algorithm and elementary data structures.	K1
CLO2	Understand the various types of linked lists and dynamic storage management.	K2
CLO3	Infer the concepts of trees, binary tree traversal and graph	K3
CLO4	Understand Static and Dynamic Symbol Tables, Hashing and apply different Hashing Functions	K3
CLO5	Apply sorting and searching techniques to organize and process data efficiently for various computational problems	K3
K1- Remember; K2 - Understand; K3 – Apply		

CLO – PLO Mapping

CLOs/PLOs	PLO1	PLO2	PLO3	PLO4	PLO5
CLO1	1	3	1	-	-
CLO2	1	3	2	1	2
CLO3	2	2	2	2	2
CLO4	1	3	2	1	2
CLO5	2	3	3	2	2
3 - Substantial (high)		2 - Moderate (medium)		1 - Slight (low)	

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Unit	Content	No. of Hours
I	Introduction: Introduction of Algorithms – Analyzing Algorithms. Abstract Data Type, Arrays: - The Sparse Matrices-Representation of Arrays. Stacks and Queues: The Stack Abstract Data Type - The Queue Abstract Data Type - Evaluation of Expressions – Infix to Postfix Conversion, Multiple Stacks and Queues	18
II	Linked List: Singly Linked List – Linked Stacks and Queues –Representing Polynomials as Singly Linked List– Polynomial Addition - Doubly Linked List and Dynamic Storage Management, More on Linked Lists – Garbage Collection and Compaction	18
III	Trees: Basic Terminology – Binary Tree Representations - Binary Trees – Traversal. Binary Search Trees: Introduction - Searching a Binary Search Tree - Inserting into a Binary Search Tree - Deletion from a Binary Search Tree - Height of a Binary Search Tree, Graphs: Terminology and Representations-Traversals- State Space Representation in AI -Shortest Paths, Transitive Closure.	18
IV	Symbol Tables: Static Tree Tables-Dynamic Tree Tables- Hash Tables: Hashing Functions -Overflow Handling, Dynamic Hashing using Directories.	18
V	Sequential Search: Binary Search- Definitions. Insertion Sort- Radix Sort, Quick Sort- 2Way Merge Sort, Heap Sort. Case Study: Analysis of various Sorting Techniques.	18
Total Hours		90

Text Book

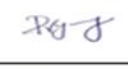
1.	Ellis Horowitz, Sartaj Sahni, Susan Anderson-Freed (1993), Fundamentals of Data Structures in C
2.	Samanta, D. (2017). Classic data structures (2nd ed.). PHI Learning.

Reference Books

1.	Mark Allen Weiss, Data Structures and Algorithm Analysis in C, 2 nd Edition, Pearson Education Asia.
2.	Ellis Horowitz, Sartaj Sahani and Dinesh Mehta (2008), Fundamentals of Data Structures in C++, 2 nd Edition, University Press.

Web Resources (Swayam / NPTEL)

1.	https://onlinecourses.swayam2.ac.in/nou24_cs06/preview
2.	https://onlinecourses.swayam2.ac.in/cec19_cs04/preview
3.	https://onlinecourses.nptel.ac.in/noc21_cs79/preview
4.	https://www.geeksforgeeks.org/artificial-intelligence/state-space-search-in-ai/

Course	Name	Signature
Prepared By	Dr.G.Priyadarshini, Faculty, Department of Computer Applications	
Vetted By	Dr.P.Vijayakumar, HoD, Department of Computer Applications	
Validated By	BoS Chairman	